

TrustGuard: The Ultimate Hackathon Q&A Defense Guide

Mastering Judge Interrogations, Technical Skepticism & Defense Strategies

Track 01 (PS-02: AI for Digital Trust) · Innovators Conclave 2026 · Evaluation Prep

JUDGES SCORING RUBRIC VULNERABILITY ALERT:

Under official evaluation rules, 15 marks are directly tied to 'Technical Depth & Code Quality' and 'Q&A Handling & Team Contribution'. An explicit deduction penalty is issued if: **(1) Only one member answers questions**, **(2) A confidence score is presented as proof**, or **(3) A single-modality model is dressed up as multimodal**. This guide equips the entire team with crisp, authoritative, synchronized responses to dominate the Q&A panel.

PART 1: The 'Doesn't This Exist?' Questions (Everyday & Market Reality)

Q1: "Wait, doesn't Grammarly or QuillBot already check writing and detect AI? How are you different?" [Recommended Speaker: Lead Presenter]

■ **The 30-Second Clincher:** Grammarly and QuillBot are essay and grammar editors—they evaluate generic text in an absolute vacuum. Grammarly checks if a comma is missing; QuillBot checks if an academic essay was generated by ChatGPT. Neither of them has any concept of who the author is, identity impersonation, financial fraud, or voice/video deepfakes.

■ **Deep-Dive Technical Backing:** Consider real-world fraud: an attacker hacks an executive's LinkedIn and writes: 'Dear team, please wire **■**25,00,000 immediately to vendor account 1234.' To Grammarly or QuillBot, that message gets a **100% Green Score** because it has perfect grammar and polite tone! TrustGuard flags it because: (1) Rajesh Verma never uses imperative financial demands, (2) The accompanying voice note exhibits neural vocoder phase artifacts, and (3) The destination bank account is not in corporate records. **Grammarly checks grammar; TrustGuard stops identity theft and multimillion-dollar wire fraud.**

■ **Trap to Avoid:** Never demean the questioner. Give the relatable example above so non-technical evaluators immediately have an 'Aha!' moment.

Q2: "Does our system already exist as it is in the commercial market?" [Recommended Speaker: Architecture / Strategy Lead]

■ **The 30-Second Clincher:** No single, unified platform exists as a holistic system today. The current market is completely fragmented into three isolated silos: (1) *Unimodal Deepfake Checkers* (Hive, Reality Defender, Deepware) that only scan uploaded video pixels; (2) *Enterprise Brand Monopolies* (ZeroFox, BrandShield) that charge \$50,000/yr for human SOC analysts; and (3) *Academic AI Detectors* (Turnitin, ZeroGPT) that only score student essays.

■ **Deep-Dive Technical Backing:** No existing tool combines: (1) Consented Personal Identity Vaults (biometric face/voice baseline under India's DPDP Act 2023), (2) An Always-On Browser Extension measuring continuous profile authenticity (0%–100%) directly in social feeds, (3) Stylometric writeprint anomaly detection against a personal life summary, and (4) Cross-modal lip-sync and cheapfake archive grounding with an actionable human verification playbook. TrustGuard is the first unified Digital Trust platform accessible to individuals and professionals.

■ **Trap to Avoid:** *Acknowledge that deepfake point-solutions exist, but emphasize that TrustGuard unifies identity, media, and context into a single place.*

Q3: "How are you different from dedicated deepfake detectors like Hive Moderation, Truepic, or Sentinel?" [Recommended Speaker: ML / Forensic Specialist]

■ **The 30-Second Clincher:** Existing deepfake checkers are isolated binary filters: they take a raw file and guess '80% AI' without knowing who is depicted, what is happening, or how to verify it. TrustGuard is an identity-anchored relational trust platform that correlates persona writeprints, cross-modal consistency, and real-world context.

■ **Deep-Dive Technical Backing:** Hive and Sentinel operate in a complete vacuum: they look at pixels, miss 100% of 'cheapfakes' (real videos paired with fake breaking news captions), and fail completely when legitimate media is compressed on WhatsApp. In contrast, TrustGuard connects the user's canonical identity baseline, checks for visual homoglyph clones on social feeds, catches lip-sync audio discrepancies, and gives human analysts an actionable verification playbook.

■ **Trap to Avoid:** *Do NOT argue that Hive or Truepic are bad companies. Say: 'They solve file-level inspection; we solve persona impersonation and relational digital trust.'*

Q4: "Why not just plug into GPT-4o or Claude's multimodal vision API and ask it if a video is real?" [Recommended Speaker: ML / Forensic Specialist]

■ **The 30-Second Clincher:** LLMs hallucinate confidence, lack biometric grounding, and cannot compute forensic frequency spectra or face/voice vector embeddings. Furthermore, relying purely on a cloud LLM costs \$0.05 per query and takes 4 to 8 seconds—unusable for a real-time browser companion.

■ **Deep-Dive Technical Backing:** TrustGuard uses a hybrid pipeline: local, deterministic extractors (OpenCV 2D-FFT in 40ms, Librosa vocoder phase analysis in 60ms, ArcFace biometrics) run on CPU instantaneously. We only use lightweight LLM reasoning at the final synthesis stage to formulate the structured JSON Dual-Polarity Ledger.

■ **Trap to Avoid:** *Emphasize that LLMs cannot compute mathematical Fourier transforms or extract biological subcutaneous pulse (rPPG).*

Q5: "Who is your actual target user, and how will this make money post-hackathon?" [Recommended

Speaker: Strategy Lead]

■ **The 30-Second Clincher:** We serve a high-value B2C and B2B hybrid model: (1) High-Risk Individuals & Creators (executives, journalists, influencers) paying \$15/mo for clone hunting and likeness defense; (2) Enterprise Fraud & HR Teams paying per-API inspection to verify executive wire requests and prevent AI candidate fraud.

■ **Deep-Dive Technical Backing:** The market need is urgent: deepfake CEO fraud and romance/crypto impersonation represent a \$10B+ global crisis. Under India's DPDP Act 2023, corporations face extreme liability if they fail to prevent impersonation fraud. TrustGuard provides compliance-ready audit trails.

■ **Trap to Avoid:** Don't say 'It's free for everyone forever.' Hackathon judges love seeing realistic B2B SaaS revenue potential.

PART 2: Technical & Algorithmic Defense

Q6: "How do you scan social media profiles without paying \$5,000/month for X APIs or getting banned?" [Recommended Speaker: Full-Stack / Extension Engineer]

■ **The 30-Second Clincher:** We completely bypass backend scraping through our Always-On Browser Extension. The extension inspects public profile DOM elements directly inside the user's active, authenticated browser session in real-time as they view the page.

■ **Deep-Dive Technical Backing:** For baseline profile creation, we use consented 'Bring Your Own Data' (BYOD): users upload their official GDPR/Takeout data archives (LinkedIn/X exports) or run the client extension on their profile. This eliminates backend server IP bans, zero API costs, and complies with platform Terms of Service.

■ **Trap to Avoid:** This is your biggest winning answer—judges expect you to be trapped by API paywalls. Your browser-client architecture is the silver bullet.

Q7: "What happens when generative AI becomes 100% photorealistic and pixel artifacts disappear entirely?" [Recommended Speaker: Team Lead / ML Specialist]

■ **The 30-Second Clincher:** That is precisely why TrustGuard was built! Unimodal pixel detectors will all become obsolete. TrustGuard wins because we do not rely solely on pixel noise—we examine relational consistency and context.

■ **Deep-Dive Technical Backing:** Even if an AI video is photorealistic, an attacker cannot fake: (1) Out-of-character writing stylometry, (2) Historical event archives (checking if the video background is from 2022), (3) Spatio-temporal whereabouts, and (4) Out-of-band cryptographic voice/face hash matching. When pixels deceive, context reveals the truth.

■ **Trap to Avoid:** Quote the problem statement verbatim: 'The question is not Is this image AI-generated? It is: Can an AI system examine multiple pieces of evidence and determine consistency?'

Q8: "Can an attacker fool your stylometry model by just prompting ChatGPT: 'Write like Person X'?" [Recommended Speaker: Stylometry / NLP Specialist]

■ **The 30-Second Clincher:** ChatGPT can mimic superficial keywords, but it cannot replicate subconscious forensic linguistic invariants: Function Word Distributions, Yule's Characteristic K (lexical concentration), and Syntactic POS bigram transitions.

■ **Deep-Dive Technical Backing:** Furthermore, LLM-generated text possesses a distinct statistical signature: uniform low perplexity and low burstiness. Real human communication is naturally bursty with idiosyncratic punctuation, sentence variation, and rhythm. When an attacker uses ChatGPT to mimic an author, the Perplexity Delta engine flags the artificial uniformity.

■ **Trap to Avoid:** *Don't get lost in mathematical formulas during Q&A; just explain that LLMs write with 'predictable smoothness' while humans write with 'erratic burstiness.'*

Q9: "What if the user is a sparse poster with only 5 tweets? How do you build an identity baseline?" [Recommended Speaker: ML Specialist]

■ **The 30-Second Clincher:** TrustGuard incorporates an Epistemic Uncertainty Estimator (U_{epistemic}). If baseline data is sparse, the system attenuates the weight of the stylometry engine and alerts the user: 'Baseline confidence is provisional due to limited text history.'

■ **Deep-Dive Technical Backing:** In sparse text cases, the system shifts weight to other modalities: external domain verification, cryptographic C2PA manifests, visual face centroid matching, and handle homoglyph analysis. We never guess when evidence is scarce.

■ **Trap to Avoid:** *Never pretend the system is 100% confident with 5 tweets. Acknowledging uncertainty shows high engineering maturity.*

Q10: "How does the system catch 'Cheapfakes' (real authentic video reused with a fake caption)?"

[Recommended Speaker: Forensic / Video Specialist]

■ **The 30-Second Clincher:** We utilize Perceptual Hashing (pHash & PDQ) coupled with reverse visual knowledge grounding. When a video is submitted, we extract keyframe perceptual hashes and query historical news and archive registries.

■ **Deep-Dive Technical Backing:** If a video circulating today claims to be 'Breaking News: Floods in Mumbai 2026', but its pHash matches a verified 2021 disaster archive video, TrustGuard immediately flags a **Contextual Repurposing Anomaly**—even though the video itself has zero AI pixel modifications.

■ **Trap to Avoid:** *Make sure to emphasize that traditional deepfake detectors score cheapfakes as '0% Fake', completely missing the attack.*

PART 3: Accuracy, Reliability & 'No Score is Proof'

Q11: "Why do you insist that 'No Score is Proof'? Doesn't that admit your AI cannot be trusted?"

[Recommended Speaker: Team Lead]

■ **The 30-Second Clincher:** Quite the opposite: admitting that 'No score is proof' is the definition of forensic rigor. In real-world law and cybersecurity, a probabilistic AI score cannot throw someone in jail or block a million-dollar corporate transaction without human verification.

■ **Deep-Dive Technical Backing:** TrustGuard replaces the scalar number with a **Calibrated 5D Trust Vector** and a **Dual-Polarity Ledger** (showing both Red Flags and Green Counter-Evidence). Most importantly, we provide an interactive **Independent Verification Playbook** (e.g. out-of-band PBX calls, archive trace) that empowers a human to verify the evidence and sign an audit certificate.

■ **Trap to Avoid:** Point to the problem statement icon: ■ **'NO SCORE IS PROOF'**. Judges put that requirement there specifically to test if teams understand uncertainty!

Q12: "What about false positives? What if a real executive is on a blurry hotel Wi-Fi connection?"

[Recommended Speaker: Forensic / Signal Specialist]

■ **The 30-Second Clincher:** A standard detector flags compression artifacts (H.264 blockiness, packet drop) as diffusion generation artifacts, causing an embarrassing false accusation. TrustGuard explicitly isolates compression through high-frequency Fourier analysis.

■ **Deep-Dive Technical Backing:** When high compression noise is detected without vocoder phase cuts or identity mismatch, the system elevates the **Epistemic Uncertainty factor ($U > 0.50$)** and caps maximum suspicion, reporting: '*Authentic / Low Confidence due to Network Compression*'. This completely prevents false alarms.

■ **Trap to Avoid:** Mention that this is demonstrated live in '*Scenario D: The Low-Bandwidth Edge Case*'.

Q13: "How does the Continuous Authenticity Score in the browser extension work?" [Recommended

Speaker: Extension / UX Specialist]

■ **The 30-Second Clincher:** Rather than a binary 'Real or Fake' flip, the extension evaluates a composite multi-factor trust polynomial (0% to 100%): (1) Handle Homoglyph Distance (Cyrillic substitutions), (2) Account Creation Age vs. Stature, (3) Avatar Reverse-Search Originality, and (4) Interaction Reciprocity.

■ **Deep-Dive Technical Backing:** It renders as a subtle, color-coded pill: ■ **90%+ (Established & Verified)**, ■ **50–80% (New Account / Needs Caution)**, and ■ **<30% (High-Risk Impersonator)**, allowing users to make informed decisions before answering DMs or clicking links.

■ **Trap to Avoid:** Show how this fits naturally into user browsing without interrupting their workflow.

PART 4: Privacy, Governance & Hackathon Scope

Q14: "How do you store sensitive biometric face and voice data without violating India's DPDP Act 2023 or GDPR?" [Recommended Speaker: Security / Architecture Lead]

- **The 30-Second Clincher:** Under India's Digital Personal Data Protection Act (DPDP 2023), mishandling biometric data carries penalties up to **■250 crore**. TrustGuard is Zero-Knowledge by Design: we NEVER store raw photos, videos, or audio recordings.
- **Deep-Dive Technical Backing:** We extract irreversible, normalized mathematical embeddings (512-d float vectors for faces, 192-d for voices). These vectors are salted, encrypted at rest using AES-256-GCM with user-derived keys, and can be permanently purged with a single click. You cannot reconstruct a human face from an ArcFace embedding.
- **Trap to Avoid:** Mentioning DPDP Act 2023 by name will impress Indian hackathon evaluators.

Q15: "What did you actually build in the 24-hour hackathon versus what is future roadmap?"
[Recommended Speaker: Team Lead]

- **The 30-Second Clincher:** We built a functional, end-to-end prototype running locally on CPU in <150ms: (1) Web Dashboard Workbench, (2) Deterministic 2D-FFT and Librosa audio spectrogram extractors, (3) Stylometric Writeprint engine, (4) Pydantic-enforced XAI reasoning synthesizer, (5) Interactive Verification Playbook with PDF export, and (6) Browser Extension UI mockups.
- **Deep-Dive Technical Backing:** What is strictly future roadmap (clearly delineated for Round 2): automated high-volume web scrapers across 50+ platforms, enterprise Kafka streaming, and hardware security module (HSM) biometric enclaves. We only committed in Round 1 to what we are delivering in Round 2.
- **Trap to Avoid:** Crucial defense: Avoids the official penalty trap 'Scope promised in R1 not delivered in R2'.

PART 5: Team Q&A Assignment Matrix

Ensure all members participate actively during panel interrogation to secure full marks (5/5).

Domain / Topic Area	Assigned Team Member	Core Key Terms to Mention
Everyday AI (Grammarly/QuillBot vs TrustGuard)	Member 1 (Lead Presenter)	Grammar vs Identity fraud, Relational trust, 'No score is proof', DPDP Act 2023
Market Comparison & Point Solutions	Member 2 (Architecture/Strategy)	Siloed tools (Hive/ZeroFox) vs Unified system, B2B SaaS, Continuous trust gauge
Media Forensics & Signal Analysis	Member 3 (ML/Signal Specialist)	2D-FFT azimuthal roll-off, Vocoder phase cuts, Lip-sync phoneme-viseme, rPPG
Stylometry & Persona Anomaly	Member 4 (NLP/Stylometry Specialist)	Writeprints, Yule's Characteristic K, Perplexity delta, Coercive urgency, POS transi
Browser Extension & Scraping Defense	Member 1 / 4 (Full-Stack/Extension)	DOM-level inspection, Zero API cost, Continuous trust gauge, AES-256 encryption